

Tools and QC

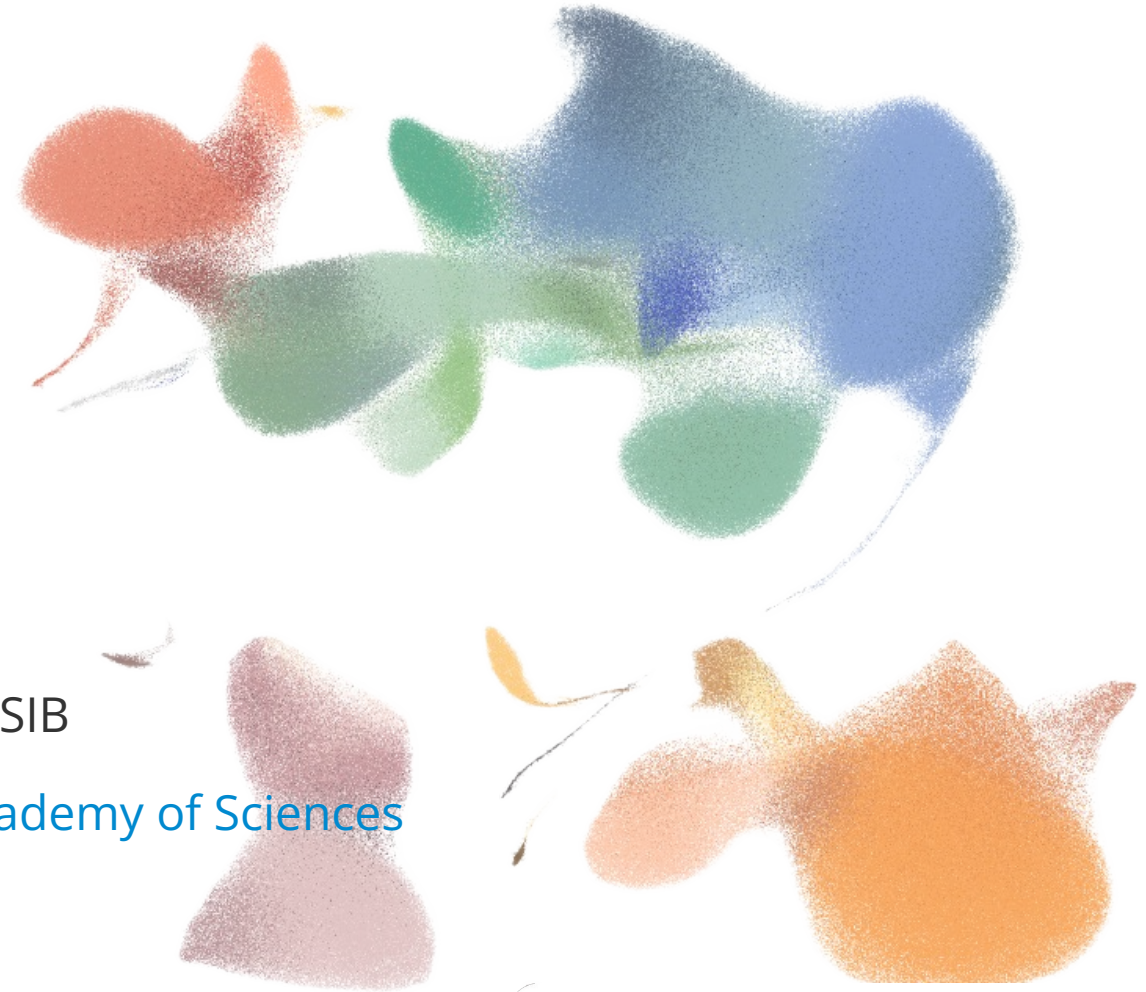
COURSE ON SCRNA-SEQ DATA ANALYSIS

Popular tools and fine filtering of data

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with acknowledgments to Deepak Tanwar and SIB

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Input data – counts

cells ->

genes ->

	ATTC-1	CCGG-1	GCGA-1
RPL22	5	13	3
PARK7	0	9	3
ENO1	1	35	0
PLA2G2A	0	0	0
CAMK2N1	0	6	0
CDC42	0	8	1
C1QA	0	0	25
C1QC	0	0	25
C1QB	0	0	29
ID3	0	35	0
RPL11	18	29	16
CLIC4	0	4	0

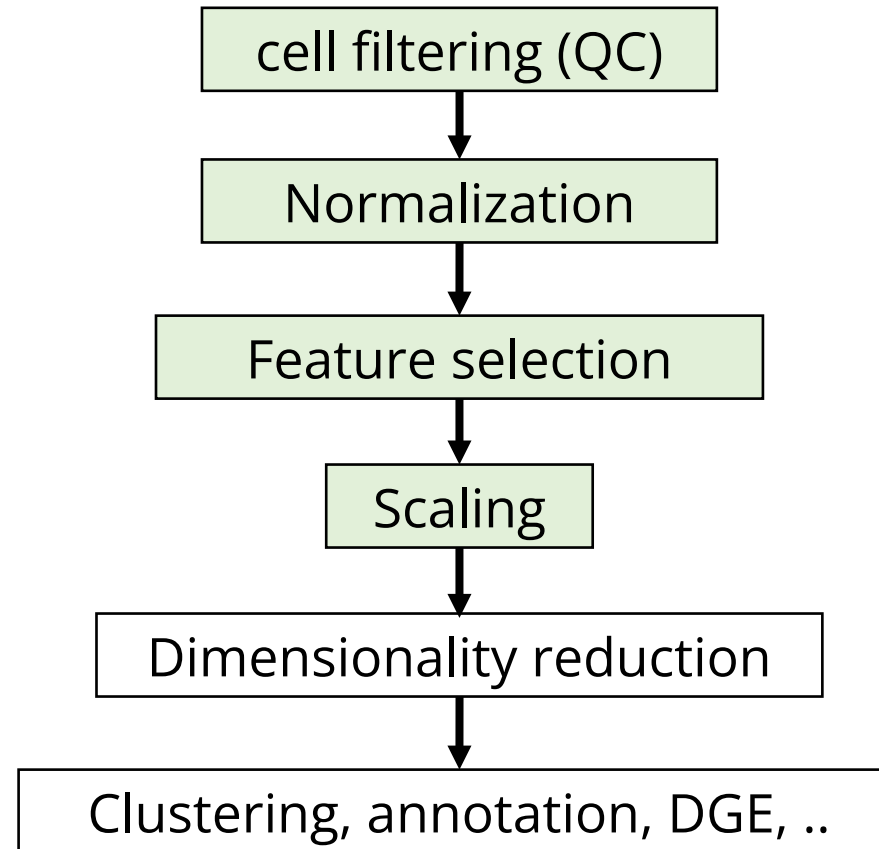
Input data – cell data

variables ->

cells ->

	Sample name	nFeature_RNA	nCount_RNA
ATTC-1	PBMMC-1	13	3
GCGA-1	PBMMC-1	9	3
AGAT-1	PBMMC-1	35	0
GGCT-1	PBMMC-2	0	0

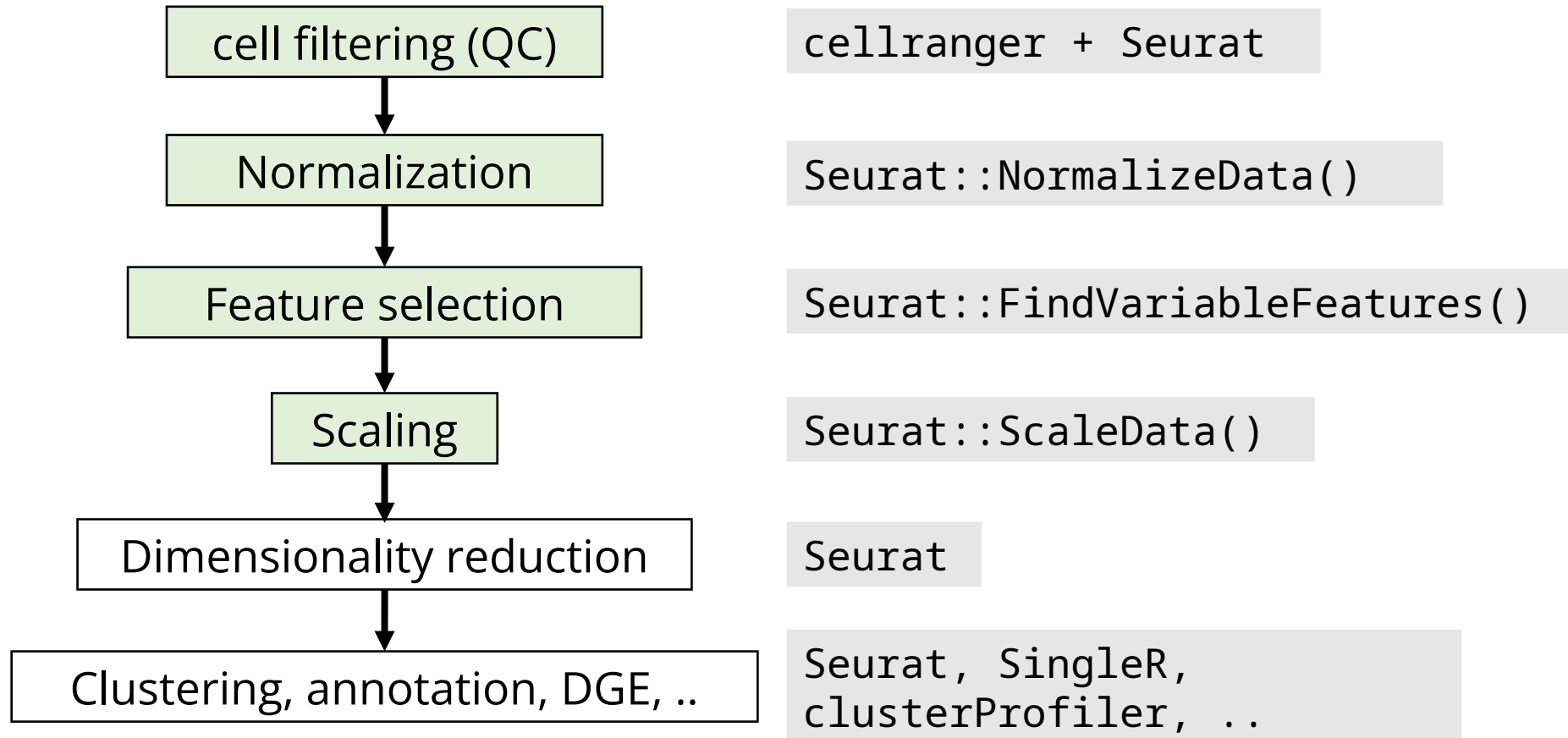
Analysis overview



Toolkits vs tools

- Toolkits (qc, normalization, reduction, plots)
 - Seurat
 - Bioconductor
 - Scanpy
- Many other specialized tools for specific tasks
 - <https://www.scrna-tools.org/>

Analysis overview



Cell filtering

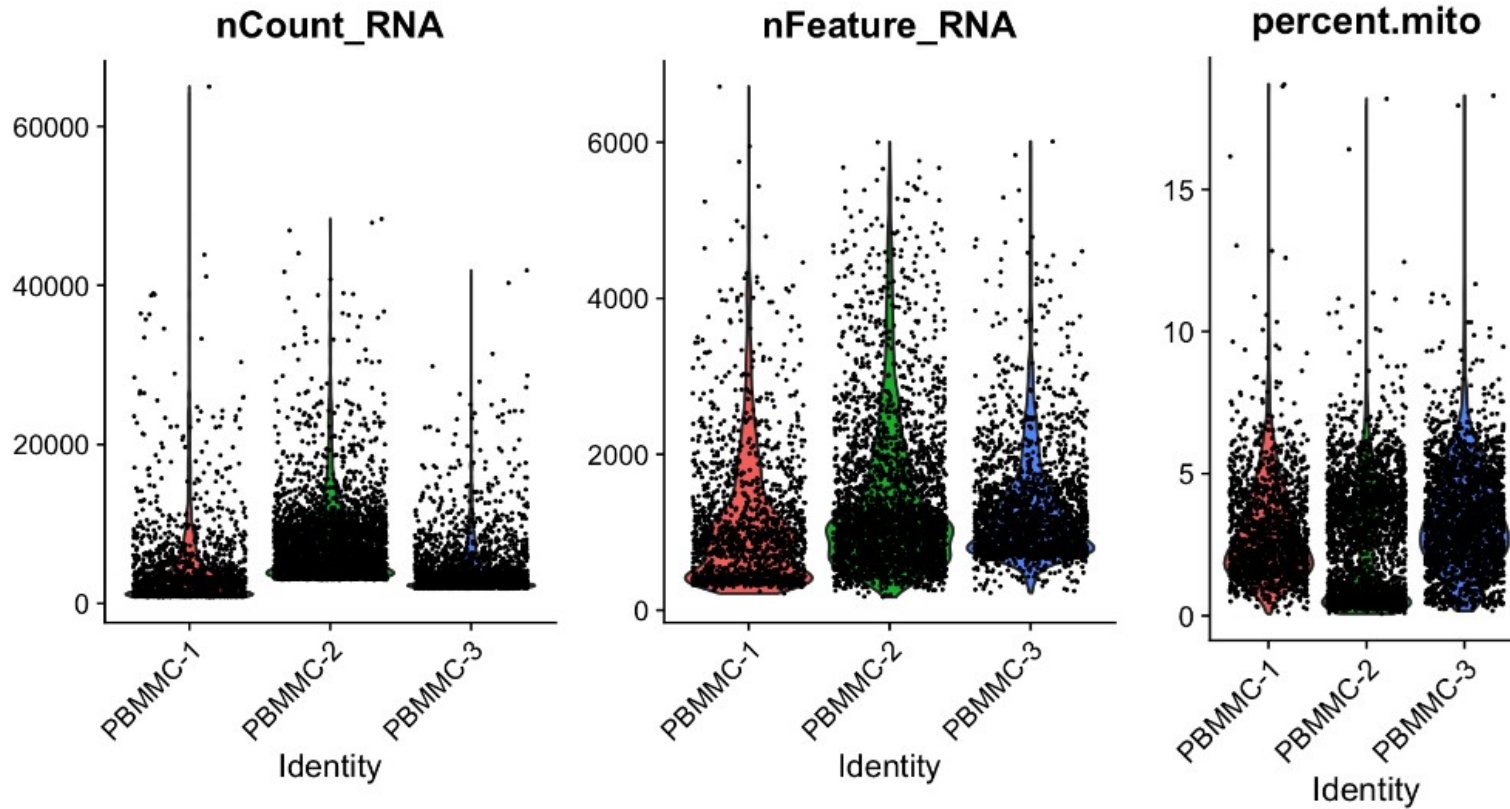
- Goal = filter count matrix with biological filters
 - Raw or filtered cellranger output
- Variables to consider
 - Total expression
 - Detected genes count
 - % mitochondrial UMI: dying cells
 - % globin or ribosomal UMI
 - Relationships between variables

Cell filtering

- Variables to consider per cell

Variable	High value	Low value	Action
Total UMI count	Doublet	Empty droplet	Filter extremes
Detected genes count	Doublet	Low quality cell	Filter extremes
% of mitochondrial UMIs	Stressed/dying cell	-	Filter high (~>10%)
% of ribosomal UMIs	Technical bias/stress	-	Filter/flag high

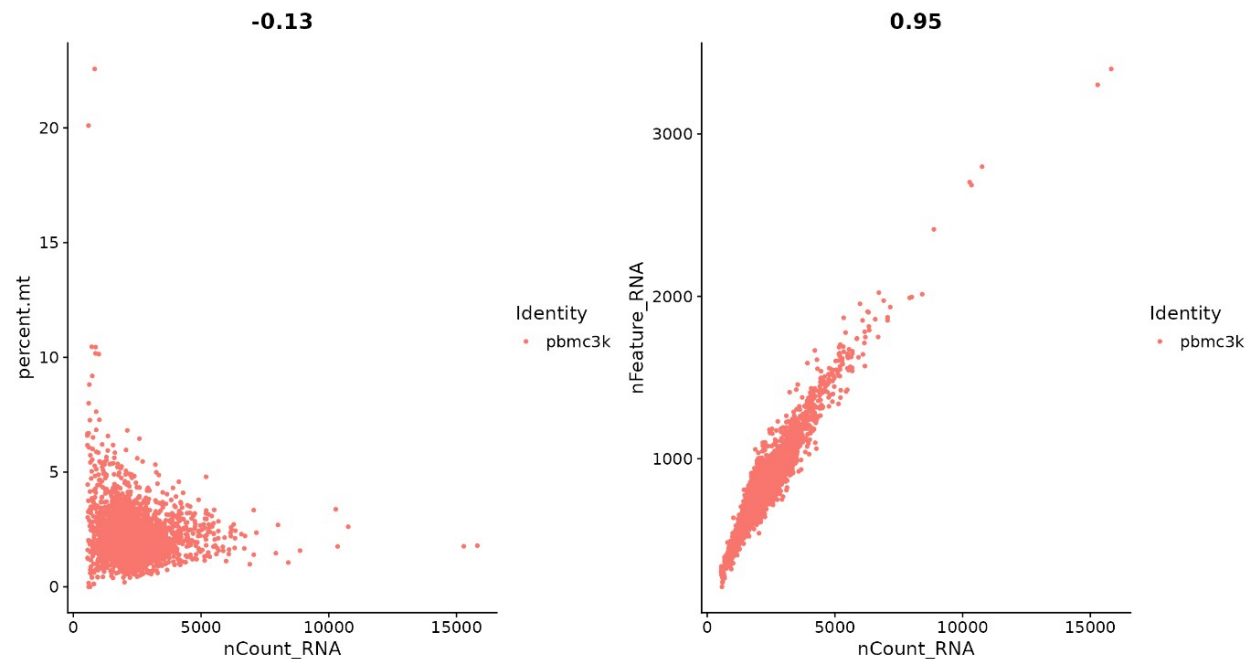
Cell filtering



Think about thresholds for each experiment

Cell filtering

- Relationships
 - High UMI ~ High gene count
 - High %MT ~ Low UMI/gene count



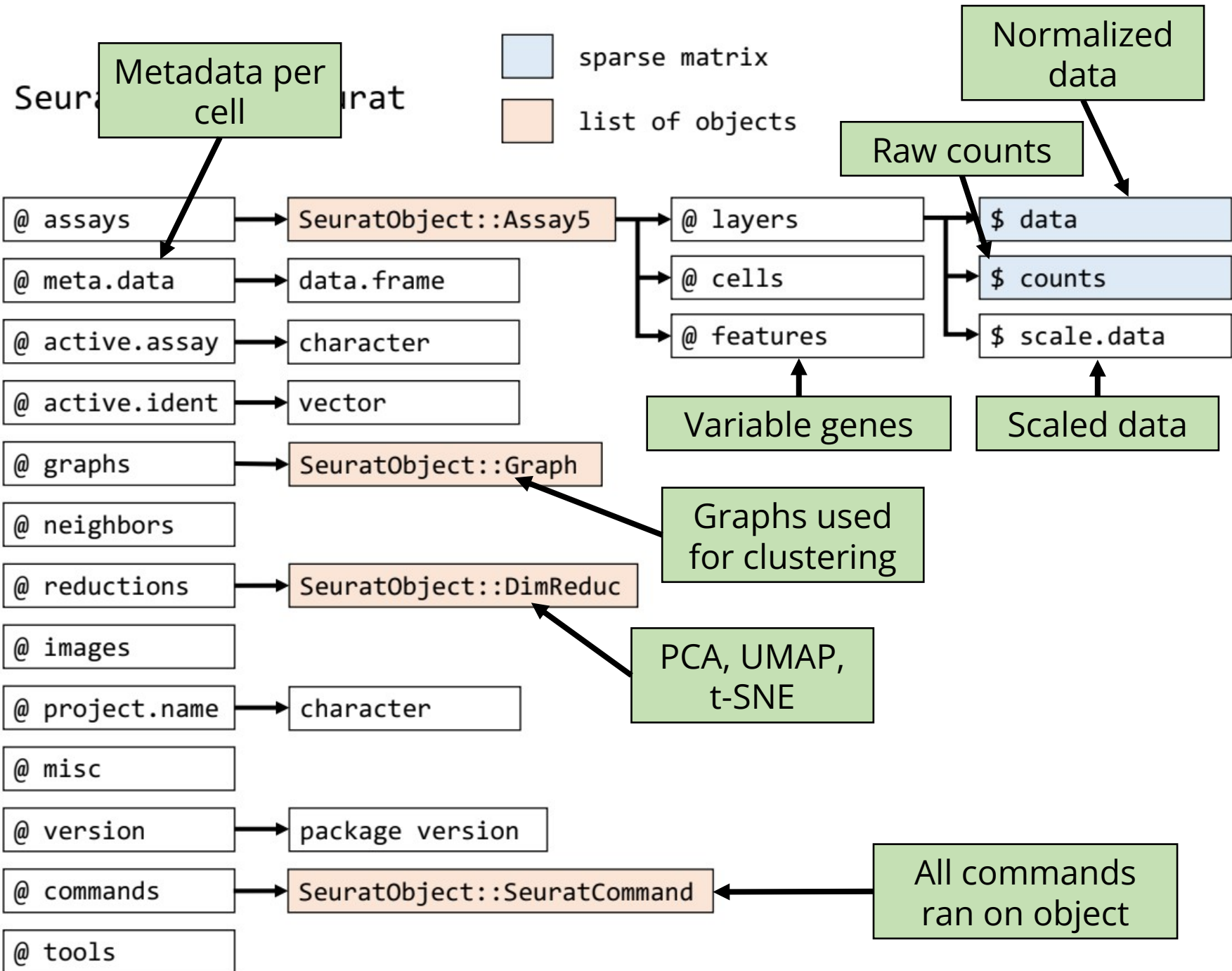
Think about thresholds for each experiment

Cell filtering

- Data sensitive approach
 - `scuttle::perCellQCFilters()` - MAD approach
 - assumes high quality data
 - observe the results

Gene filtering

- ?



Thank you for your attention!